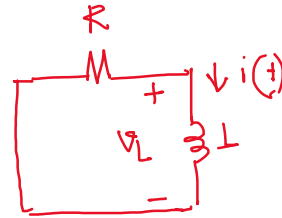
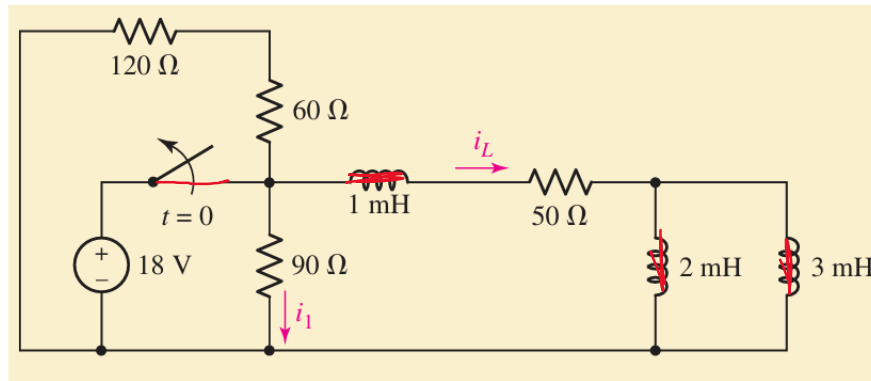


Tuesday, September 2, 2025 10:16 AM

Source free: R, L 

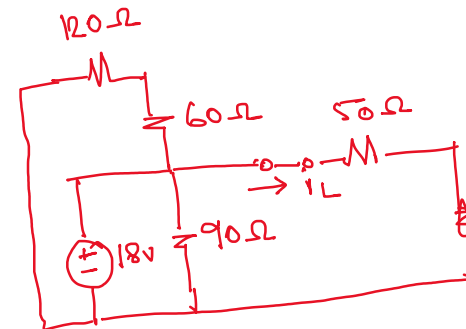
$$i(t) = I_0 e^{-t/\tau}$$

$$\tau = \frac{L}{R}$$



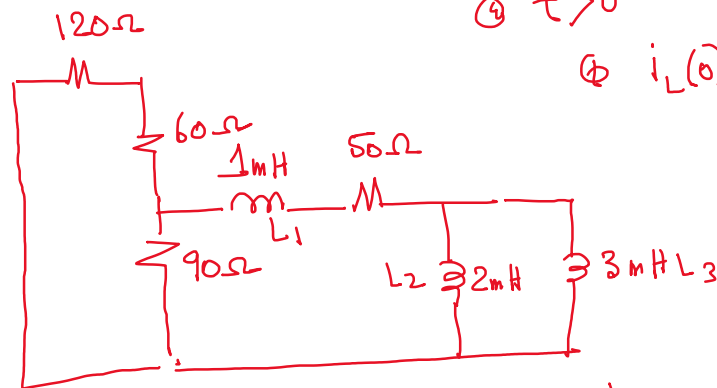
@ $t < 0$
 i_L

$t < 0$.
Inductor short ckt



@ $t > 0$

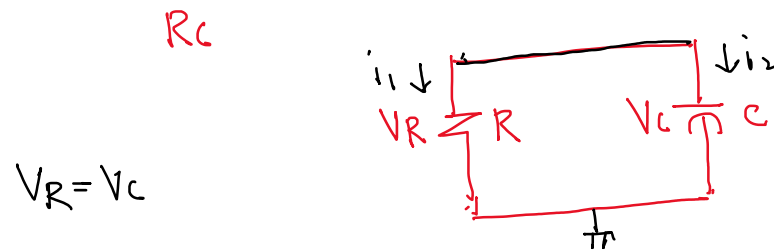
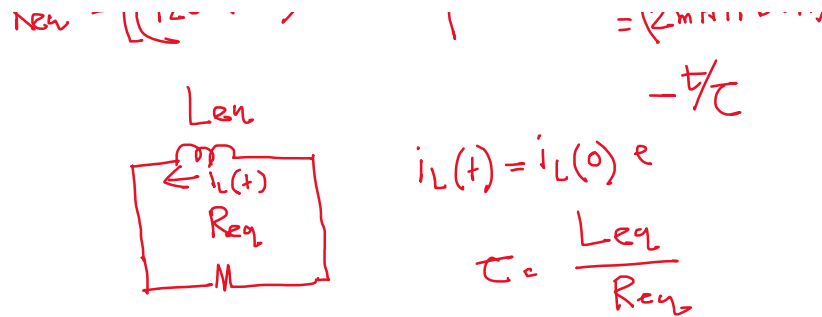
$i_L(0) = i_L$ For $t < 0$



$$\tau = \frac{L}{R} = \frac{1}{\frac{1}{120+60} + \frac{1}{90}} + 50$$

$$L_{eq} = (L_2 || L_3) + L_1$$

$$= (2mH || 3mH) + 1mH$$



$$\frac{di}{dt} + \frac{R}{L} i = 0$$

$$i(t) = I_0 e^{-\frac{R}{L}t}$$

$$= i(0) e^{-t/\tau}$$

Source Free

$$\frac{V_R}{R} + C \frac{dV_C}{dt} = 0$$

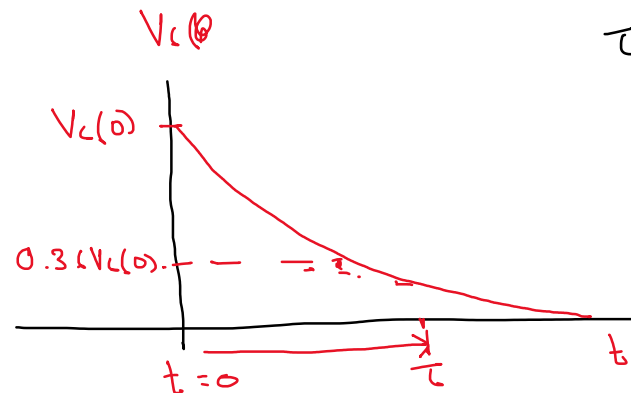
$$\Rightarrow \frac{V_C}{R} + C \frac{dV_C}{dt} = 0$$

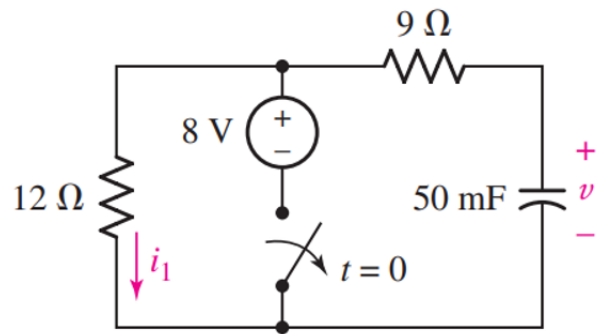
$$\Rightarrow \frac{dV_C(t)}{dt} + \frac{V_C(t)}{RC} = 0$$

$$V_C(t) = V_C(0) e^{-\frac{t}{RC}}$$

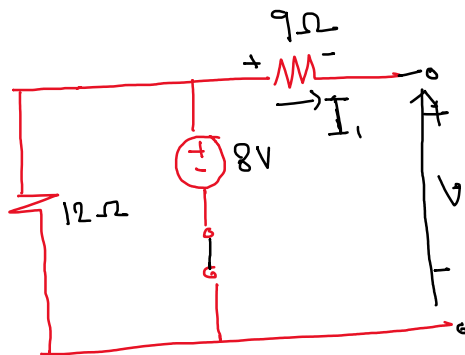
$$V_C(t) = V_C(0) e^{-t/\tau}$$

$$\tau = RC$$





to DC signal, capacitor is open

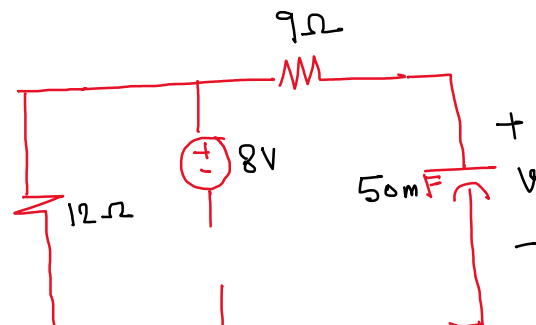


$$+8 - 9I_1 - v = 0$$

$\Rightarrow V = 8V$, voltage across the Cap for $t < 0$

$t = 0$, the switch is open.

$$@ t = 0, V(0) = 8V$$

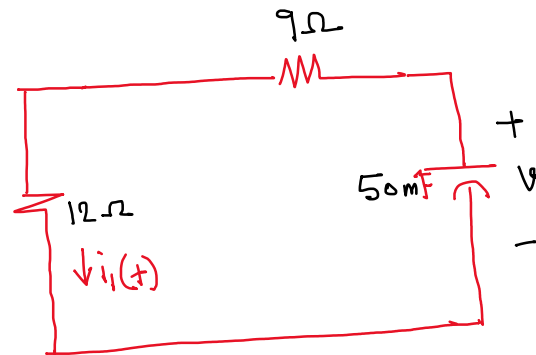


Voltage across the cap doesn't change instantaneously

$t = 0^-$ [Instant before the switch was open]

$t = 0^+$ [Instant after the switch is open]

$$V(0^+) = V(0^-) = 8V$$



$$R_{eq} = 9 + 12 = 21 \Omega$$

$$C = 50 \text{ mF}$$

$$\tau = R_{eq}C = 21 \times 50 \times 10^{-3} \text{ sec}$$

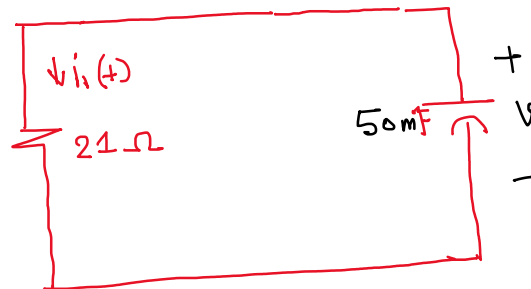
→

$$-t/\tau$$

$$V(t) = V(0^+) e^{-t/21 \times 50 \times 10^{-3}}$$

$$= 8 e^{-t/\tau}$$

$$i_1(t) = \frac{V(t)}{R_{eq}} = \frac{8 e^{-t/\tau}}{21} \text{ Amp.}$$



$$\tau = R_{eq}C$$

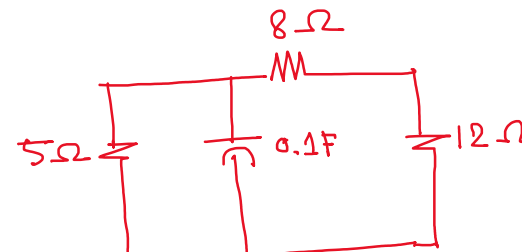
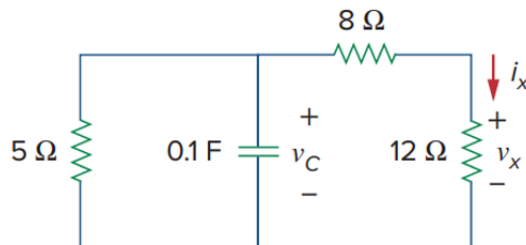


Figure 7.5
For Example 7.1.

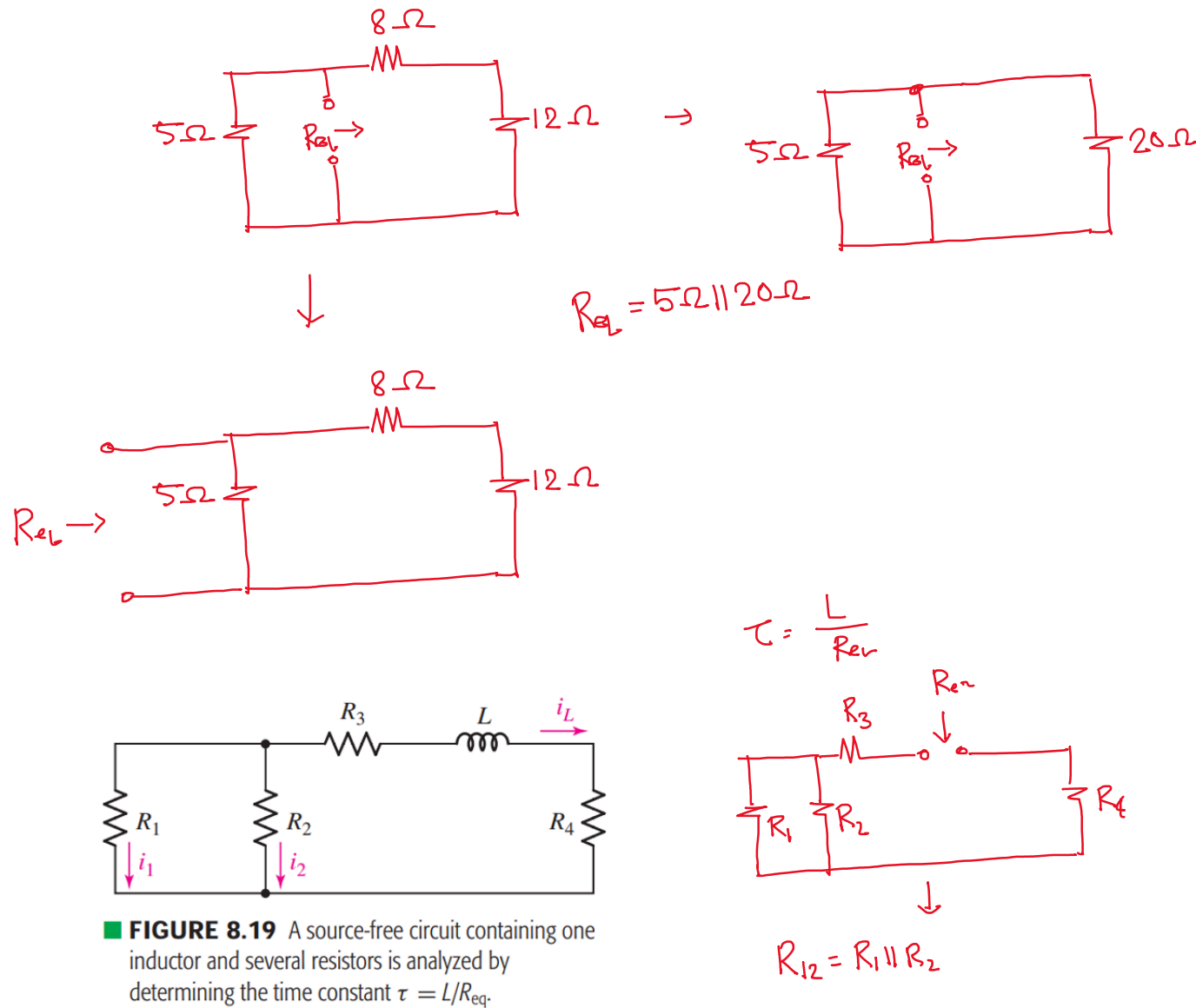
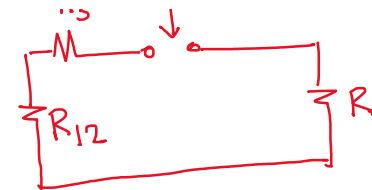
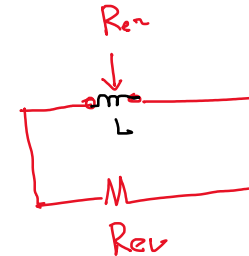


FIGURE 8.19 A source-free circuit containing one inductor and several resistors is analyzed by determining the time constant $\tau = L/R_{\text{eq}}$.



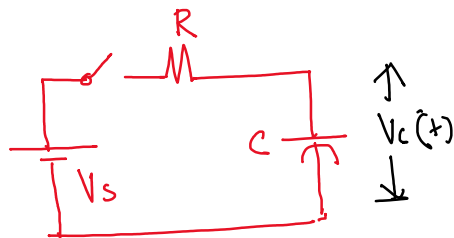
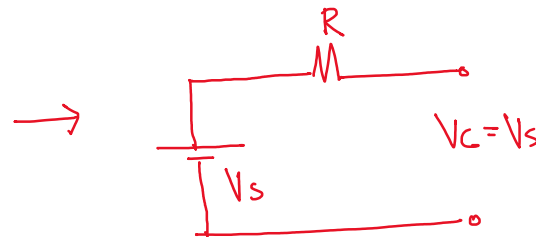
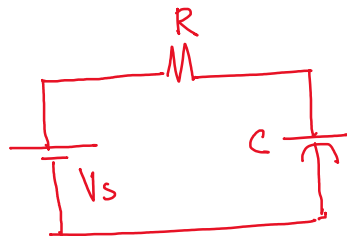
$$R_{eq} = R_3 + R_{12} + R_4$$

$$\tau = \frac{L}{R_{eq}}$$



Cap is open ckt to dc signal at steady state

Source free



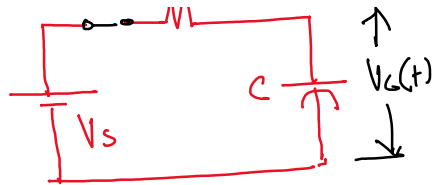
② $t < 0$, Switch is open for a long time

③ $t = 0$, the switch is suddenly closed

Fan. $t < 0$, its at (0 rpm)

④ $t = 0$, you turned the switch on

$$t = 0 \quad R$$



0rpm $\rightarrow$ 300rpm $\rightarrow$ stay at 500V"
 ' transient + steady state
 (changing with time) (unchanged with time)

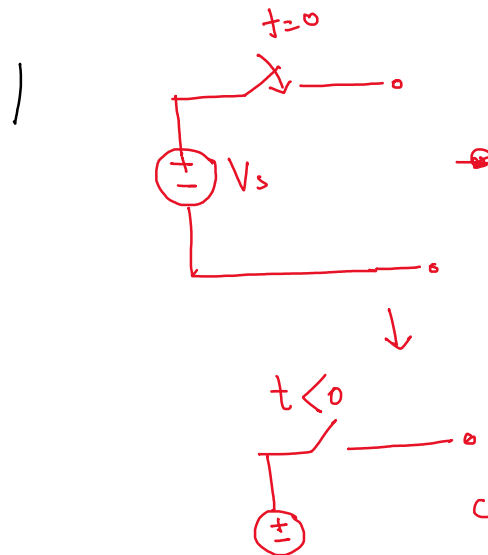
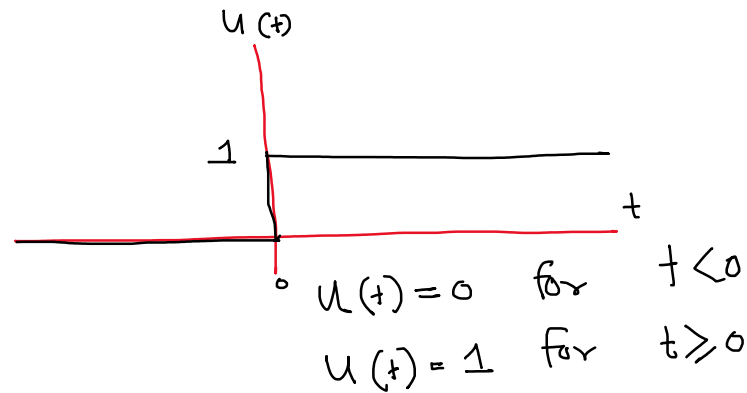
Forced Response

$$t < 0, V_s = 0$$

$$t \geq 0, V_s = V$$

unit step function
 EEE 243

$u(t)$



V

$V_s u(t)$

